NAN LI

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EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) Sep. 2025 – Present
Master of Philosophy in Artificial Intelligence *CGA: 4.00/4.30*

- Supervisors: Prof. Yutao Yue (primary) and Prof. Bojun Cheng.

Guangzhou University Sep. 2021 – Jun. 2025
Bachelor of Science in Physics *GPA: 3.86/4.00; Rank: 3/98*

RESEARCH INTERESTS

Egocentric Multimodal Learning; Reliable & Efficient Visual Reasoning; Human-Centered AI.

PUBLICATIONS & MANUSCRIPTS

[1] **PAVE: Preference-Aligned Visual Egocentric Learning for the Blind and Low-Vision Users**
Nan Li, Shaofeng Liang, Shiyun Zhao, et al.
Under review at WWW 2027 (CCF-A) *First Author*

[2] **One Direction, Many Routes: Uncovering a Shared Adversarial Vulnerability in Dynamic-Depth Language Models**
Shiyun Zhao*, Shaofeng Liang*, **Nan Li***, et al.
Under review at WWW 2027 (CCF-A) *Equal Contribution*

[3] **ObsBridge: Observation-Grounded Conditioning for Ego Trajectory Prediction**
Zhenhao Wang, Shaofeng Liang, Jiemin Wu, Sijia Chen, Keni Shi, **Nan Li**, et al.
Under review at CVPR 2027 (CCF-A) *Co-author*

[4] **DRMOT: A Dataset and Framework for RGBD Referring Multi-Object Tracking**
Sijia Chen, Shaofeng Liang, Lijuan Ma, Yanqiu Yu, **Nan Li**, et al.
Under review at IEEE TCSVT (CCF-B; JCR Q1; IF: 10.8) *Co-author*

[5] **All-Optical Neuromorphic Logic Processing Units Using Monolithic Black Phosphorus**
Huide Wang, Yi Liu, Wenkai Wang, Ziqian Wang, Yihan Zhu, Yanqi Ge, Songrui Wei, Bowen Du, Meng Qiu, **Nan Li**, et al.
Under review at Science Advances (JCR Q1; IF: 13.9) *Co-author*

RESEARCH EXPERIENCE

Preference-Aligned Egocentric Visual Assistance [1] Jun. 2026 – Oct. 2026
HKUST(GZ)

- Led dataset construction, method development, and evaluation for blind and low-vision (BLV) assistance; built PA-EgoQA (4,762 questions from 1,302 videos) with six BLV-oriented response criteria and video-grounded human review.
- Developed question-conditioned response policies for single-pass VLM inference, combining structured supervised fine-tuning with correctness-first Filter-GRPO to improve user-oriented responses while maintaining factual accuracy.
- Evaluated three VLM backbones; improved InternVL2.5-8B accuracy from 62.88% to 67.00% and judge-estimated BLV Score from 3.07 to 3.29 (out of 5), while reducing response length by 37.8%.

Adversarial Robustness in Dynamic-Depth LLMs [2] Jul. 2026 – Oct. 2026
HKUST(GZ)

- Investigated embedding-space adversarial attacks and cross-architecture transfer across SkipGPT, DiffSkip, and four Router-Tuning configurations on ten zero-shot benchmarks.
- Demonstrated cross-routing vulnerability, observing a 42.9% relative accuracy drop on the source model and 33.3–35.3% drops across unseen routing configurations.

RGB-D Referring Multi-Object Tracking [4] Jan. 2026 – May 2026
HKUST(GZ)

- Refined and polished figures and tables for the DRMOT manuscript, improving visual clarity, consistency, and the presentation of experimental results.

Future-Proofing STEM Education: An AI Tutoring Agent Oct. 2025 – Present
HKUST(GZ)

- Co-developed an AI tutoring agent for undergraduate physics, integrating conversational tutoring, memory, front-end interaction, and multimodal student-state perception.
- Independently designed and implemented BERR, a four-frame eyeglass-glare removal network with degradation-aware temporal fusion; constructed 2,243 synthetic glare bursts for robust facial expression recognition.
- On 2,824 paired RAF-DB faces under strong synthetic glare, improved frozen POSTER V2 FER accuracy by 2.94 percentage points and Macro-F1 by 3.16 points over direct FER; integrated the BERR–FER pipeline into the tutoring platform.

RecurBool: Boolean Circuit Compilation Toward Photonic Recurrent Computing Apr. 2026 – Present
HKUST(GZ) / SZU

- Designed a precision-aware framework for compiling trained GRU language models into fixed two-input Boolean circuits, preserving gated memory updates and autonomous recurrent state feedback.
- Developed groupwise quantization, per-dimension codebook calibration, and error compensation for 5-bit weights and recurrent states; verified bit-exact agreement with finite-precision references and near-baseline performance versus original FP32 GRUs on text8 (1.7436 vs. 1.7116 BPC) and WikiText-2 (1.9165 vs. 1.8903 BPB).

All-Optical Neuromorphic Logic Processing Units [5] Dec. 2025 – Aug. 2026
HKUST(GZ) / SZU

- Designed and trained a differentiable Logic-SNN for a collaborative photonic neuromorphic computing study, integrating learnable Boolean logic gates with Leaky Integrate-and-Fire (LIF) neurons for spiking neural computation.
- Implemented soft-to-hard logic-gate conversion and spiking evaluation; achieved 98.05% soft and 97.54% hard accuracy on MNIST at four timesteps, and examined discretization challenges on CIFAR-10.

RESEARCH APPOINTMENT

Shenzhen University, Shenzhen, China Oct. 2025 – Present
Visiting Student *Hosted by Prof. Han Zhang (Optica Fellow)*

- Modeled Li-intercalated black phosphorus (Li-BP) nonlinear optical responses as differentiable activation functions; compared fixed and learnable variants in MLP/CNN simulations.
- Simulated optical neural networks; proposed an input-adaptive channel-attention architecture combining free-space diffractive optics with VCSOA cross-gain modulation (XGM), with physical experiment feasibility under investigation.
- Investigated photoluminescent perovskite–phosphor materials for optical reservoir computing, including write/read dynamics, fading memory, and neural readout.

HONORS & AWARDS

Postgraduate Studentship (MPhil), HKUST(GZ)	2025 – 2027
Chinese Undergraduate Astronomical Innovation Contest – Second Prize	2024
“Challenge Cup” China College Students’ Entrepreneurship Competition – Bronze Award	2023
“Internet+” China College Students’ Innovation and Entrepreneurship Competition – Bronze Award	2022
Guangzhou University Scholarships – First Class, Second Class	2021 – 2025

TECHNICAL SKILLS

Programming	Python, MATLAB, Bash.
Deep Learning	PyTorch, Supervised Fine-Tuning, Reinforcement Learning, Model Evaluation.
Tools	Linux, Git, LaTeX, Blender.